Inside the Atom: Protons, Neutrons and Electrons

Subatomic particles, atomic and mass numbers, ions and isotopes

Name _____ Class _____ Date _____

ATOMIC NUMBERS

H = 1 O = 8 F = 9 Al = 13 S = 16 K = 19 Fe = 26 Zn = 30

A. The particles

1. For the proton, neutron and electron, give the relative charge, the approximate relative mass, and where each is found in the atom.

2. What does the atomic number tell you, and what does the mass number tell you?

B. Counting particles

3. Complete the number of protons, neutrons and electrons for each particle:

(a) $^{19}_{9}\mathrm{F}$ (b) $^{39}_{19}\mathrm{K}^{+}$ (c) $^{32}_{16}\mathrm{S}^{2-}$ (d) $^{27}_{13}\mathrm{Al}^{3+}$ (e) $^{64}_{30}\mathrm{Zn}$

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4. Write the full symbol (with mass number, atomic number and charge) for a particle with:

(a) 26 protons, 30 neutrons and 24 electrons

(b) 8 protons, 10 neutrons and 10 electrons

C. Isotopes

5. Hydrogen has three isotopes: hydrogen-1, hydrogen-2 and hydrogen-3. How many neutrons does each have?

6. Why do isotopes of the same element have essentially the same chemical properties?

D. Challenge

7. The atomic mass of chlorine on the periodic table is 35.45, but no chlorine atom has a mass number of 35.45. Explain.

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